

ALEXANDER LLEWELLYN SMITH

La Jolla, California · afllewellynsmith@gmail.com

<https://alexllewellynsmith.netlify.app/>

<https://www.linkedin.com/in/alexander-llewellyn-smith/>

EDUCATION

Dartmouth College, Hanover, NH

June 2028

Bachelor of Engineering: Mechanical Engineering

Bachelor of Arts: Engineering Sciences, Minor: French (native speaker)

Relevant Coursework: Distributed Systems, Thermodynamics, Embedded Systems, Computer-Aided Design & Kinematics, Digital Electronics, Solid Mechanics, Machine Engineering

Activities: Dartmouth Formula Racing (Mechanical Team), Drummer in 7 Women (band), Club Snowboard, Chi Gamma Epsilon Fraternity (Exec Board, Programmer)

RELEVANT EXPERIENCE

Teaching Assistant, Dartmouth College

March 2026 – Present

ENGS 33 - Solid Mechanics (Fall 2026), ENGS 29 - Computer-Aided Design & Kinematics (Spring 2026)

- Organized office hours teaching the fundamentals of solid mechanics, including truss and bridge analysis, mechanics of materials, buckling, and beam bending, as well as SolidWorks fundamentals, and basics of 2D and 3D engineering drawings
- Ran labs for students analyzing material properties using Instron machine.
- Supported students in designing, iterating, and building scale bridges for end-of-term project
- Prepared students for CSWA (Certified SolidWorks Associate) and CSWP (Certified SolidWorks Professional) exams

Commonlands LLC, San Diego, CA

June 2025 – August 2025

Engineering Intern

- Measured lenses using Trioptics MTF (Modulation Transfer Function) machine to develop detailed product specification sheets for customers
- Reduced cycle time by a factor of 5 through design and manufacturing of custom 3D printed mounts, saving 5 hours of labor per order while maintaining product quality
- Developed code and configured a Raspberry Pi computer to display a live camera feed designed to mount focused lenses on cameras to meet customer specifications. 3D printed custom camera fixtures to ensure a consistent product
- Collaborated with head engineer to setup and run the first automated lens manufacturing machine of its kind in the U.S.

Laboratory for Computational Cellular Mechanobiology - Professor Rangamani and Dr. Francis,

May 2023 – July 2024

Department of Mechanical and Aerospace Engineering, UC San Diego

Intern

- Created 3D models to solve differential equations describing cell functions. Developed “zero-dimensional” simulations for simplified models of calcium density in cells and created research documentation still used to onboard new hires.
- Collaboratively worked on the development of open-source python software SMART, Spatial Modeling Algorithms for Reactions and Transport, which uses finite-element modeling to simulate how molecules move and react within realistically shaped cells, applied to biological processes like calcium signaling and energy production in cells.

PROJECTS

Bipedal Robot

June 2025

Solo-Project

- Designed/built a fully functional RC bipedal robot integrating mechanical kinematics with electrical systems in 3D CAD
- Optimized the footpath of a custom Watt six-bar linkage using SolidWorks motion study to create the smallest overall design possible

Wooden Scale Bridge

August 2026

Project Lead

- Benchmarked bridge designs on SAP2000, iterating on each prior design to optimize bridge strength, while maintaining light weight shape and minimizing total deflection
- Utilized novel final design to create SolidWorks assembly for FEA (Finite Element Analysis) and iteratively redesigned bridge based on maximum stress, buckling, and deflection
- Manufactured bridge utilising laser cutting, producing most accurate deflection prediction (0.171 mm), second lowest weight in the class (306.9 g), and held a max load of 4210 N before failure, determined through testing in Instron machine

SKILLS & INTERESTS

3D + 2D CAD & Design: SolidWorks CSWP, Autodesk Fusion360, Autodesk Inventor, SAP2000, GD&T, FEA, Motion Study

Mechanical: FDM + SLA 3D Printing, Laser Cutting, CNC Milling, CNC Lathe

Electrical & Software: Python, Java, C, MATLAB, LaTeX, Embedded Systems, VHDL, RTL Circuit Design, Wiring

Interests: Cars, blink-182, college football, playing drums, snowboarding, road trips, rom-coms

Languages: English (Native), French (Native), Spanish (Working proficiency)